

**IET, DAVV 1<sup>st</sup> Test      Oct 2023**  
**Applied Physics 2APRC2      (E&Tc-B and E&I branch)**

Time: 70 min

Max. Marks: 20

**NOTE: All questions are compulsory. Out of two parts in each question attempt any one part**

- Q1** I (a) Explain the formation of Newton's rings and derive an expression which shows that the radii of the dark rings of reflected light are proportional to the root of natural number. 8
- (b) What are the similarities and dissimilarities between zone plate and convex lens? Find the radii of the 1<sup>st</sup> transparent zones of a zone plate whose 1<sup>st</sup> focal length is 1m of wavelength 5893Å. 5

**OR**

- II (a) Explain the Fraunhofer diffraction effect at a single slit. Derive an expression for the condition of maxima and minima. 8
- (b) What is the role of the compensatory glass plate in Michelson's interferometer? In Michelson's interferometer if a movable mirror is moved through a distance 0.1732mm, 500 fringes crossed the field of view. Find the wavelength of light. 5
- Q2** I (a) Explain the Huygens's theory of double refraction in uniaxial crystal. 5
- (b) What is a quarter-wave plate? Calculate the thickness of a quarter wave plate if the refractive indices of E-ray and O-ray are 1.553 and 1.544 respectively. (Given  $\lambda = 5000 \text{ Å}$ ). 2

**OR**

- II (a) Describe the construction and working of Nicol prism. 5
- (b) What is optical activity? Calculate the polarization angle for water ( $\mu = 1.55$ ) 2
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